BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA49109
Course Name: PHP and MySQL Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: This course aims to provide a comprehensive grasp of PHP programming, covering syntax, variables, loops, HTML integration, string manipulation, functions, arrays, complex database operations and dynamic web applications, focusing on user interactions and database functionalities.

Pre-requisite(s): Knowledge in basic programming.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Develop and evaluate basic PHP programs integrated with HTML.
CO2: Analyze and evaluate PHP functions and string operations, focusing on recursion, string manipulation, and the use of library functions.
CO3: Plan and evaluate PHP array management techniques, encompassing both index-based and associative arrays with control statements.
CO4: Analyze and simulate PHP techniques for managing HTML forms, including data processing, file operations, and form redirection.
CO5: Develop and evaluate a PHP-MySQL project demonstrating database connectivity, basic SQL operations, and exception handling.

Module 1: Introduction to PHP **[12H]**

Demonstrate simple task (such as: to display 'hello') and basic PHP Syntax (includes defining variable, constant, data type and basic arithmetic operations with operators). demonstrate condition/decision making and loop statement. Mixing of decision and looping statements with HTML script.

Module 2: Function and String in PHP **[8H]**

Call by value, call by reference and recursive function. String creating and accessing, string searching & replacing string, formatting string and demonstrating string related library function.

Module 3: Array in PHP **[12H]**

Creating index based and associative array. Also demonstrate accessing the array elements using loop, associative array using for each () loop and few related library function. Array by including decision and loop statement.

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Module 4: Handling Html Form with PHP **[12H]**

Demonstrate to capture user inputs through form. Generating file uploaded form and redirecting a form after submission, Handling file & directory related operations (opening, closing, copying, renaming and deleting a file and directory).

Module 5: PHP MySQL Connectivity and Exception handling **[16H]**

Performing database connectivity and other basic database oriented operations. A mini project based learning of PHP MySQL database connectivity with basic database operation (DML, like Insert, delete, update, Select) and setting query parameter, executing query- join with capability of exception and error handling.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K6,K5
2.	CO2	M2	K4,K5
3.	CO3	M3	K4,K5
4.	CO4	M4	K4,K6
5.	CO5	M5	K5, K6

Text books:

1. PHP and MySQL Web Development, Luke Welling, Laura Thomson, Pearson,5th Edition.

Reference Books:

1. Programming PHP: Creating Dynamic Web Pages, Tatroe K., MacIntyre P., & Lerdorf, R., O'Reilly Press 2013, 3rd edition.